

INDIA AIR POLLUTION ANALYSIS DASHBOARD

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Abstract

Air pollution is a major environmental and health concern in India. This project presents an interactive India Air Quality Analysis Dashboard developed using Microsoft Power BI. The dashboard analyzes important air pollutants such as PM2.5, PM10, CO, NO₂, SO₂, NH₃, and Ozone, along with the overall Air Quality Index (AQI) across different states and cities. The data is cleaned, transformed, and categorized to produce meaningful visual insights using charts, maps, and KPI indicators. The dashboard helps in identifying highly polluted regions, understanding pollution levels, and comparing air quality across locations. This study demonstrates how Power BI can be effectively used for environmental monitoring, awareness, and better decision-making.

1. INTRODUCTION

Air pollution has become a serious environmental and health problem in India. Due to rapid urbanization, industrial growth, and increasing vehicle usage, the air quality in many cities has worsened over time. Large amounts of air quality data are now available from government sources, but this data is often difficult to understand in raw form. Tools like Microsoft Power BI help convert this data into easy-to-read and interactive dashboards. This project focuses on an India Air Quality Analysis Dashboard developed using Power BI. The dashboard presents pollution levels of major pollutants such as PM2.5, PM10, NO₂, SO₂, CO, NH₃, and Ozone, along with the Air Quality Index (AQI) across different states and cities. The purpose of this study is to show how visual dashboards can help people better understand pollution levels and support informed environmental decisions.

1.1 Statement of the Problem

Although a large amount of air quality data is available from government sources, it is often scattered across multiple files and difficult to analyze in its raw form. This makes it hard for students, researchers, and decision-makers to quickly understand pollution conditions and identify highly polluted areas. Most people cannot easily interpret raw numerical data to assess health risks. The main problem addressed in this project is the lack of a simple, visual, and interactive system that can show air pollution trends clearly. This project solves the problem by creating a single Power BI dashboard that brings together all important air quality information in one place for easy monitoring and analysis.

1.2 Research Questions

This study attempts to answer the following questions:

- Which states and cities in India have the highest AQI values?
- Which pollutant contributes the most to air pollution in different regions?
- How are cities distributed across different AQI

categories such as Good, Moderate, and Severe? • Which cities are the most polluted and which are the cleanest?

- How does PM2.5 compare with PM10 across cities and states?

1.3 Objectives of the Study

The main objective of this study is to analyze and visualize air quality data of India in a clear and meaningful way using Power BI. The study aims to identify major pollution trends, categorize air quality levels, and compare pollution across different states and cities. By converting large datasets into interactive visuals, the dashboard helps in understanding pollution severity and health risks. The project also aims to support awareness and data-driven decision-making for environmental protection.

1.4 Significance of the Study

This study is important because it helps in understanding the current air pollution condition in India using visual analysis. By identifying highly polluted cities and dominant pollutants, the study supports better awareness of environmental risks. The findings can help students, researchers, and policymakers understand pollution patterns and take preventive actions. It also highlights the role of data analytics tools like Power BI in environmental monitoring and public health awareness.

2. Literature Review

2.1 Global Trends in Air Quality Monitoring and Analytic

Recent studies show that air pollution monitoring has become very important across the world due to its strong impact on human health and the environment. Many countries now use digital tools and data analytics to track air quality in real time. Advanced technologies like sensors, satellite data, and business intelligence tools are widely used to analyze pollution levels and identify polluted regions. Data visualization dashboards are increasingly used to present complex air quality data in a simple and understandable way for public awareness and policy-making.

2.2 Factors Influencing Air Pollution Levels

Several factors are responsible for air pollution, such as vehicle emissions, industrial activities, burning of fuels, construction dust, and weather conditions. Population density and urban growth also play an important role in increasing pollution levels. Studies have shown that pollutants like PM2.5 and PM10 are the most harmful to human health. Seasonal changes, wind speed, and temperature also influence the spread of air pollutants. Understanding these factors helps in identifying the main sources of pollution and controlling them effectively.

2.3 Theoretical Framework

This study is based on the Decision Support System (DSS) theory, which explains how data and analytical tools support

better decision-making. By using dashboards and visual analytics, complex environmental data can be transformed into useful insights for government authorities, researchers, and the general public. The study also supports the idea that data and information systems are important resources for managing environmental problems and improving public health awareness.

2.4 Role of Data Analytics in Environmental Monitoring

The use of data analytics and visualization tools in environmental studies has increased rapidly in recent years. Interactive dashboards help users analyze pollution trends, identify high-risk areas, and compare air quality across different regions. Unlike traditional reports, visual dashboards provide real-time insights and allow users to explore the data in a more meaningful way. Studies show that tools like Power BI improve data understanding, support faster analysis, and help in effective environmental planning.

2.5 Research Gap

Although many studies focus on air pollution and AQI analysis, very few projects use business intelligence tools like Power BI to create a complete, interactive, and user-friendly air quality monitoring system for India. Most existing studies are either numerical or static in nature. This project fills this gap by developing a dynamic Power BI dashboard that visually presents air quality data, pollution categories, and regional comparisons in a single analytical framework.

3. Methodology

3.1 Research Design

This study follows a quantitative and descriptive research design based on secondary air quality data collected from government sources. The purpose of this research is to analyze air pollution levels across different states and cities in India using numerical data and visual analysis. The study focuses on understanding pollution patterns, identifying highly polluted regions, and analyzing the behavior of different air pollutants using charts, maps, and dashboards in Power BI.

3.2 Dataset Description

The dataset used in this project contains real air quality data of India collected from official government data portals. It includes important attributes such as State, City, Station, Latitude, Longitude, Pollutant Name (PM2.5, PM10, CO, NO₂, SO₂, NH₃, Ozone), and Pollution Values. The dataset also includes calculated fields such as Air Quality Index (AQI) and AQI Category. This dataset allows pollution analysis from multiple angles such as city-wise comparison, state-wise comparison, and pollutant-wise comparison.

3.3 Tools and Materials

The main tool used for data processing, analysis, and visualization is Microsoft Power BI Desktop. Power BI was used to clean the data, perform transformations, create calculated columns, and build interactive dashboards. Various visual components such as bar charts, column charts, donut charts, line charts, maps, KPI cards, and slicers were used to present the air quality information in a clear and user-friendly manner.

3.4 Data Modelling

A simple and efficient data model was created using a central air quality table supported by location-related columns such as State and City. Calculated columns were created to categorize pollution levels using AQI categories like Good, Moderate, Poor, and Severe. Relationships between location and pollution values were maintained to support interactive filtering. Slicers were used to allow users to filter data by State, City, and AQI Category for dynamic analysis.

3.5 Data Analysis Techniques

Different data analysis techniques were applied in this project, such as trend analysis, category-based analysis, geographical analysis, and comparative analysis. The study examined pollution levels across cities, compared air quality between different states, analyzed dominant pollutants, and observed AQI category distribution. These techniques helped in identifying highly polluted areas, understanding pollution severity, and presenting meaningful insights for environmental awareness and decision-making.

4. Results / Findings

4.1 Dashboard KPI Summary

The Air Quality Dashboard provides a clear overall view of pollution levels across different states and cities of India. The dashboard shows the average AQI of the country, along with the average PM2.5, PM10, and other major pollutants. The results indicate that several cities fall under the Moderate to Poor AQI category, showing unhealthy air conditions in many regions. A few cities also appear in the Severe category, indicating serious health risks. These key indicators give a quick and effective summary of India's current air quality condition and help identify regions that require immediate attention.



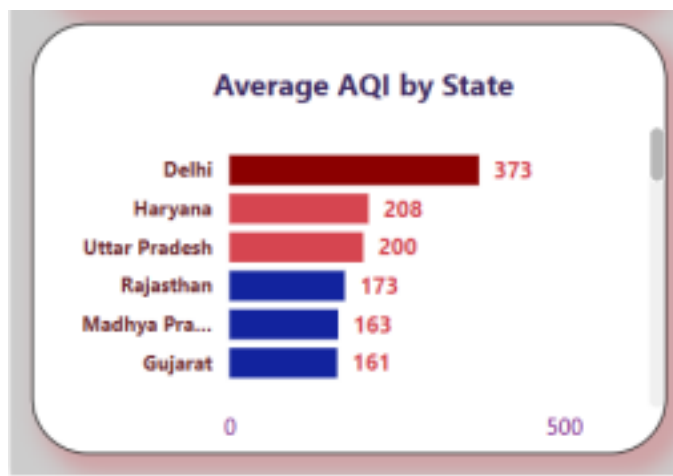
2.Key Graphical Discoveries

(a) State-wise Pollution Analysis

The state-wise AQI analysis shows that some states consistently record higher AQI values, indicating worse air quality. Highly populated and industrial regions show higher pollution levels, while some smaller or less-industrialized states show comparatively better air quality. This confirms that urbanization and industrial growth play a major role in air pollution.

Summary:

Some states show consistently high AQI levels, indicating serious pollution problems.

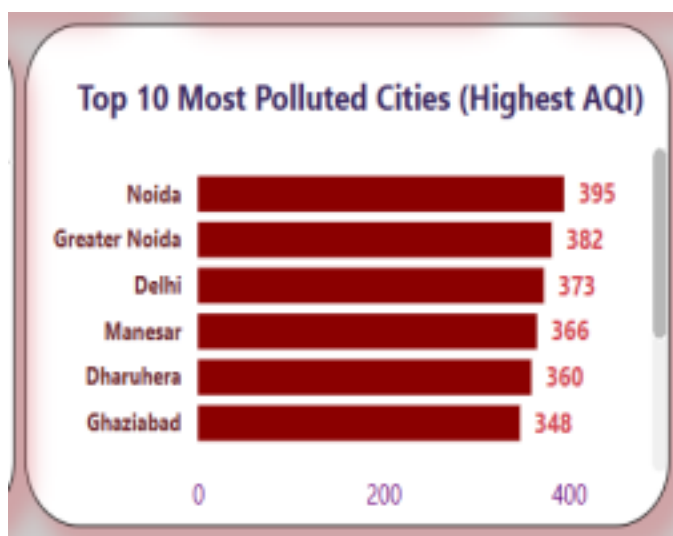


(b) City-wise Pollution Distribution

The city-level analysis highlights that a few cities appear repeatedly in the Top 10 Most Polluted Cities list. These cities mostly fall under Poor, Very Poor, or Severe AQI categories. On the other hand, a few cities also appear in the Top 10 Cleanest Cities, indicating healthier air quality. This comparison clearly shows the uneven distribution of air pollution across Indian cities.

Summary:

Certain cities suffer from extreme air pollution, while a few others enjoy relatively clean air.

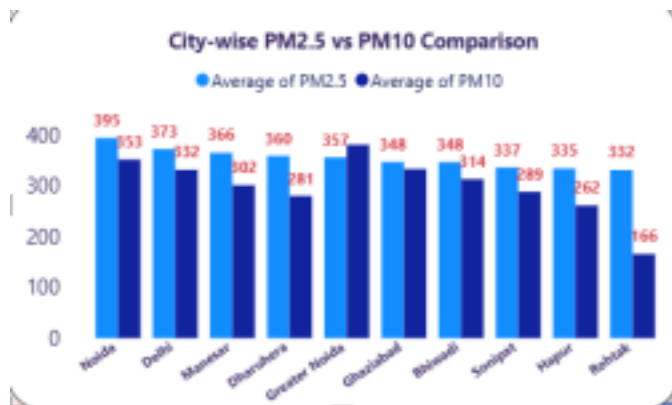


(c) PM2.5 and PM10 Analysis

The comparison between PM2.5 and PM10 shows that PM2.5 levels are generally very high in most polluted cities, making it the most dangerous pollutant. PM2.5 is more harmful because it can easily enter the lungs and bloodstream. PM10 levels are also high in some regions but are slightly less dangerous compared to PM2.5.

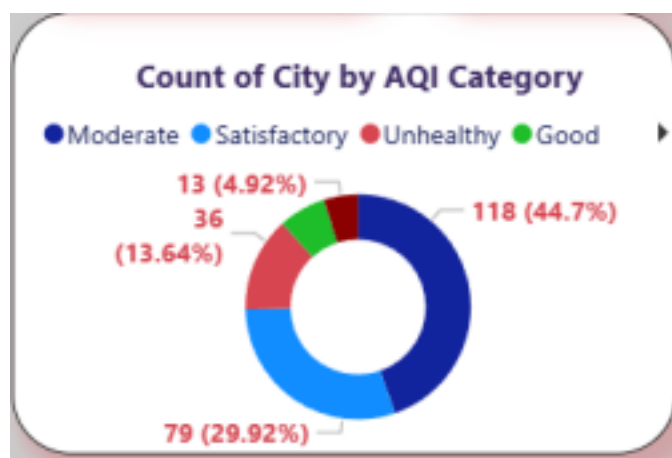
Summary:

PM2.5 is the most harmful and dominant pollutant in most Indian cities



(d) AQI Category Distribution

The AQI Category distribution shows that a large number of cities fall under the Moderate and Poor categories, while a smaller number of cities fall under the Good category. A few cities also fall under the Very Poor and Severe categories, which indicates extremely unhealthy air quality.



Most Indian cities fall under unsafe AQI categories, raising serious health concerns.

(e) Pollutant-wise Contribution

The pollutant-wise analysis shows that PM2.5 and PM10 contribute the most to overall air pollution, followed by NO₂ and CO. Pollutants like SO₂, NH₃, and Ozone also contribute but at slightly lower levels in most regions. This shows that vehicle emissions, dust, and industrial activities are the major causes of pollution.

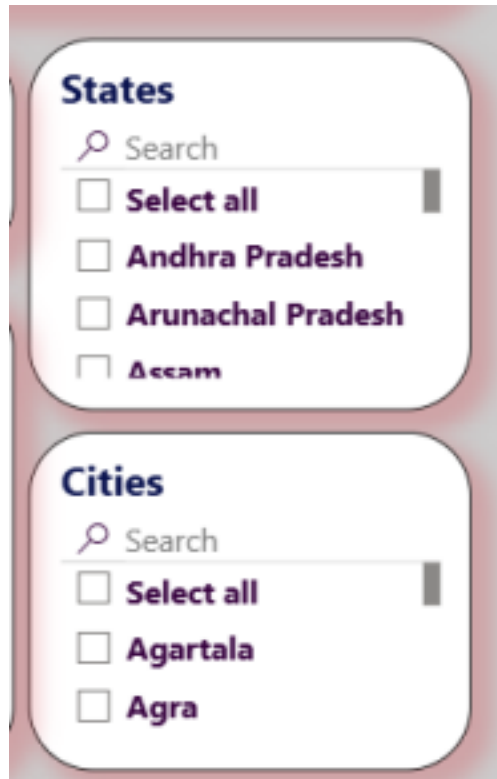
State	Count of City	Average PM2.5	Average PM10	Average NO ₂	Average CO
Andhra Pradesh	5	133.00	97.92	22.13	29.78
Assamachal Pradesh	1	124.00	108.00	6.90	12.00
Assam	6	54.93	59.80	13.47	29.45
Bihar	24	94.45	88.49	12.06	48.19
Chandigarh	1	119.00	96.50	58.00	48.50
Chhattisgarh	8	68.67	84.60	30.56	24.10
Delhi	1	373.49	332.41	96.85	75.47
Gujarat	7	127.55	137.34	27.71	31.20
Haryana	24	229.05	157.28	28.82	49.30
Himachal Pradesh	1	88.00	112.00	34.00	55.00
Jammu and Kashmir	1	132.00	130.00		16.00
Total	264	193.20	117.41	28.56	37.86

PM2.5 and PM10 are the main contributors to poor air quality.

(f) Interactive Dashboard Slicers

The dashboard includes interactive slicers such as State, City,

and AQI Category, which allow users to filter and analyze specific regions. These slicers make it easy to focus on a particular state or city and understand its pollution condition without changing the entire dashboard view.



Slicers allow fast and easy location-based air quality analysis.

5. Discussion

5.1 Interpretation of Air Quality Trends

The results from the dashboard show that air pollution levels vary widely across different states and cities of India. Many highly populated and industrial regions show high AQI values, indicating unhealthy air conditions. The dominance of pollutants like PM_{2.5} and PM₁₀ clearly shows that dust, vehicle emissions, and industrial smoke are the main causes of air pollution. The presence of several cities in the Poor and Severe AQI categories highlights the seriousness of the pollution problem and its direct impact on public health.

5.2 Influence of Urbanization and Environmental Factors

The analysis shows that air pollution is more severe in large cities and industrial areas. Factors such as heavy traffic, construction activities, factory emissions, and population density play a major role in increasing pollution levels. Weather conditions like temperature and wind also affect the spread of pollutants. Cities with fewer industries and less traffic generally show better air quality. These results prove that both human activity and environmental conditions strongly influence air pollution levels.

5.3 Comparison with Previous Studies

The findings of this project are consistent with earlier research that identifies PM_{2.5} as the most dangerous and dominant pollutant in urban areas. Previous studies also show that pollution levels are usually higher in densely populated regions. The results from this dashboard match these patterns,

which confirms that the analysis is reliable and realistic. The use of Power BI helped in clearly visualizing these patterns, making the results easier to understand and verify.

5.4 Implications

The findings of this study are important for government authorities, environmental agencies, and the general public. Cities with high AQI values should be given priority for pollution control measures. Strict monitoring of industries, control on vehicle emissions, and promotion of clean energy are necessary to improve air quality. The dashboard can help policymakers track pollution levels in real time and take rapid action. It can also increase public awareness and encourage people to follow eco-friendly practices.

6. Conclusion

This project successfully demonstrates how Power BI can be used to analyze and visualize air quality data of India in a clear and interactive way. The dashboard helps in identifying highly polluted states and cities, understanding the contribution of different pollutants, and analyzing AQI categories. The results show that PM_{2.5} and PM₁₀ are the most harmful pollutants, and many cities still fall under unsafe AQI levels. The project proves that data visualization plays an important role in environmental monitoring, public awareness, and decision-making. Overall, this study supports the effective use of business intelligence tools for improving environmental understanding and protecting public health.

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